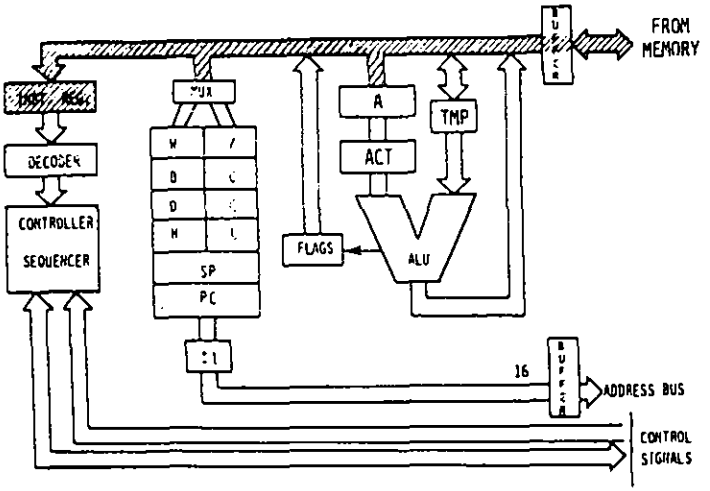

Fig. 2.19: The Instruction Arrives from the Memory into IR

It should be noted that state T4 of M1 will always be required. Once the instruction has been deposited into IR during T3, it is necessary to *decode* and *execute* it. This will require at least one machine state, T4.

A few instructions require an extra state of M1 (state T5). It will be skipped by the processor for most instructions. Whenever the execution of an instruction requires more than M1, i.e., M1, M2 or more cycles, the transition will be directly from state T4 of M1 into state T1 of M2. Let us examine an example. The detailed internal sequencing for each example is shown in the tables of Figure 2.27. As these tables have not been released for the Z80, the 8080 tables are used instead. They provide an in-depth understanding of instruction execution.

LD D, C

This corresponds to MOV r1, r2 for the 8080. Refer to line 1 of Fig. 2.27.

By coincidence, the destination register in this example happens to be named "D". The transfer is illustrated in Figure 2.20.

This instruction has been described in the previous section. It transfers the contents of register C, denoted by "C", into register D.

The first three states of cycle M1 are used to fetch the instruction from the memory. At the end of T3, the instruction is in IR, the Instruction Register, from which point it can be decoded (see Figure 2.19).

During T4: (S S S) ► TMP.

The contents of C are deposited into TMP (See Figure 2.21).